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(54) **ELECTRONIC DEVICE, CONTROL
METHOD OF ELECTRONIC DEVICE, AND
NON-TRANSITORY COMPUTER READABLE
MEDIUM**

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ABSTRACT

An electronic device according to the present invention includes: a processor; and a memory storing a program which, when executed by the processor, causes the electronic device to execute acquisition processing of acquiring information on orientation of an imaging apparatus during photography of an image; and execute control processing of controlling, based on the information, whether to perform tilt correction on the image to bring a position corresponding to a horizontal direction during the photography closer to a position corresponding to a front direction.

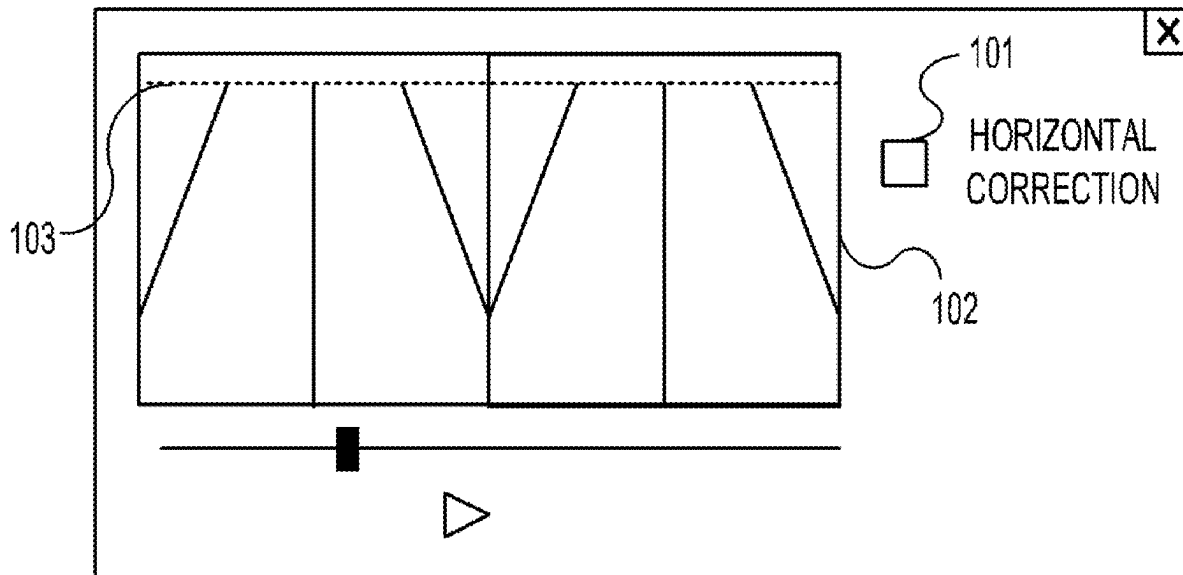


FIG. 1A

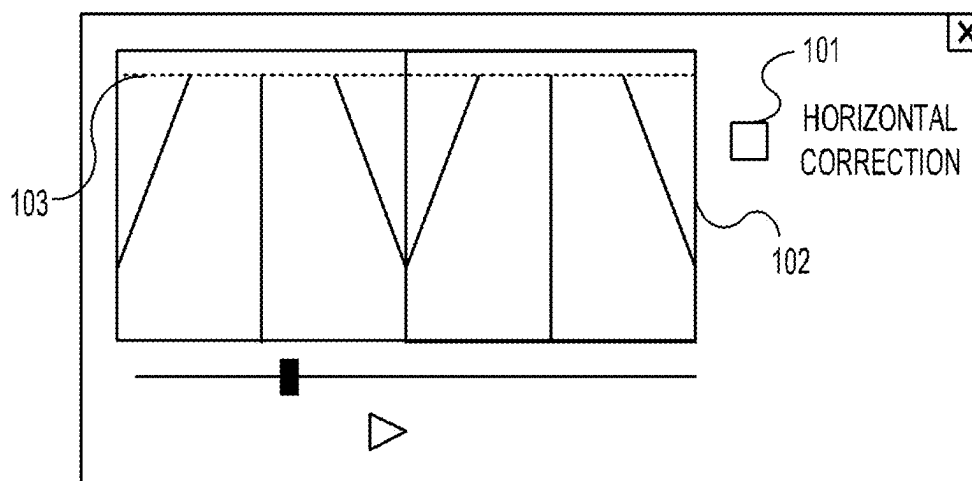


FIG. 1B

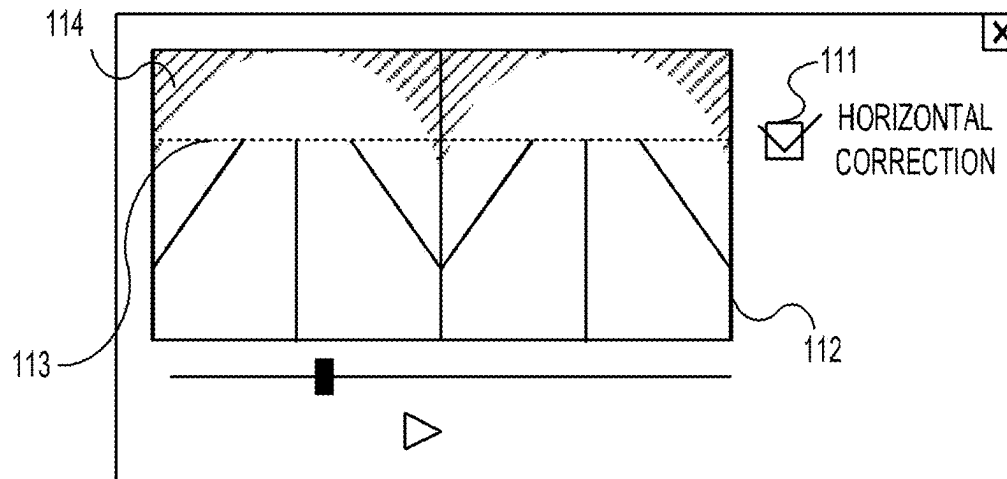


FIG. 2

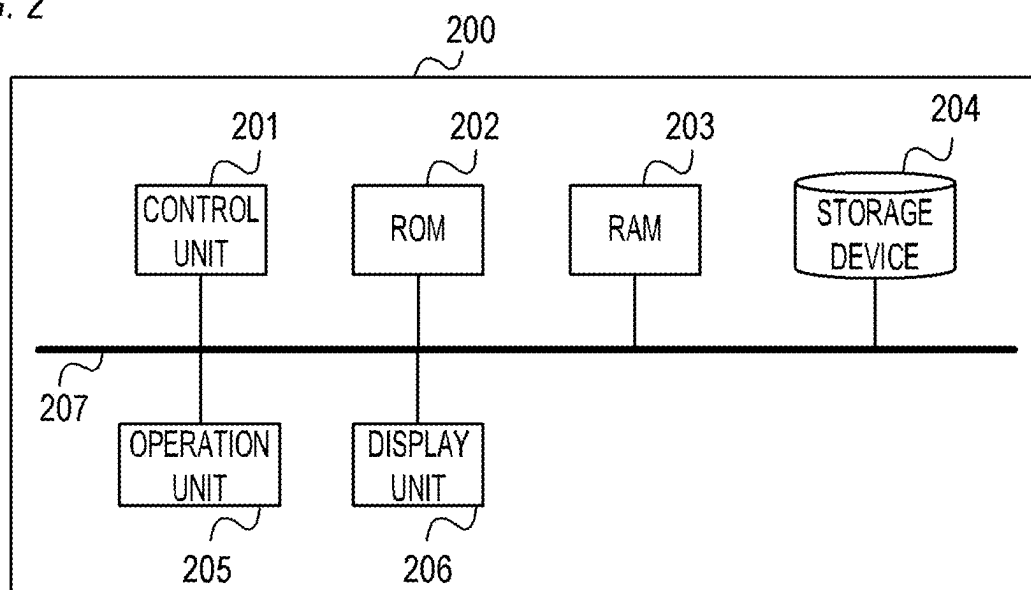
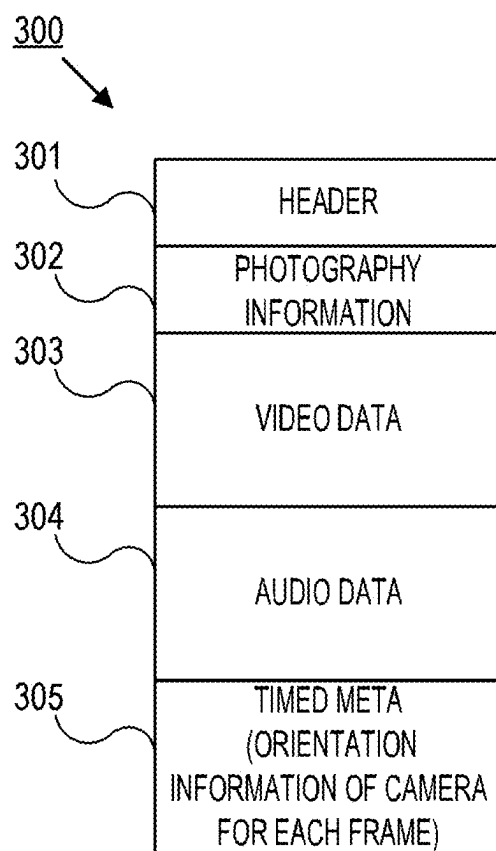


FIG. 3



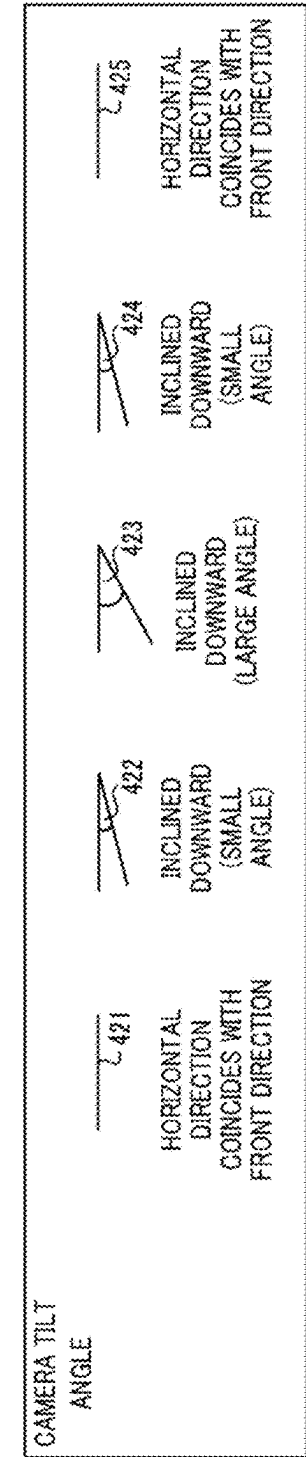
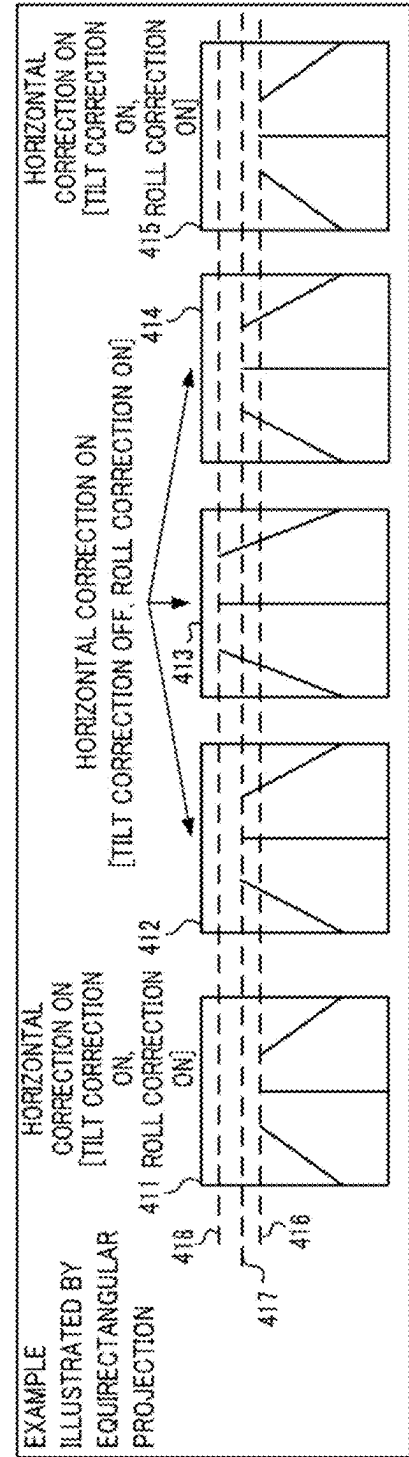
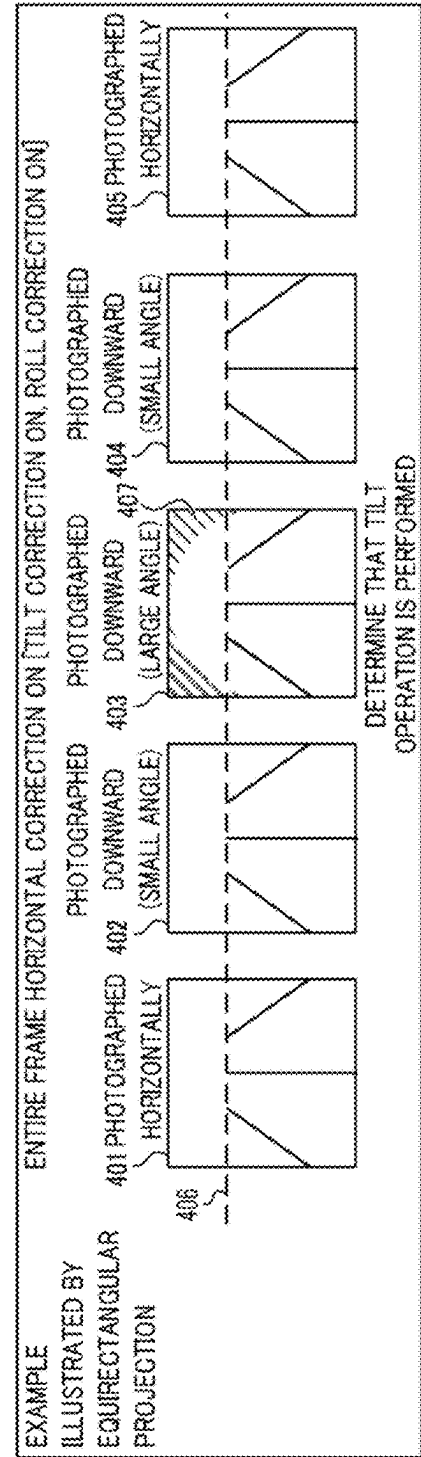


FIG. 5

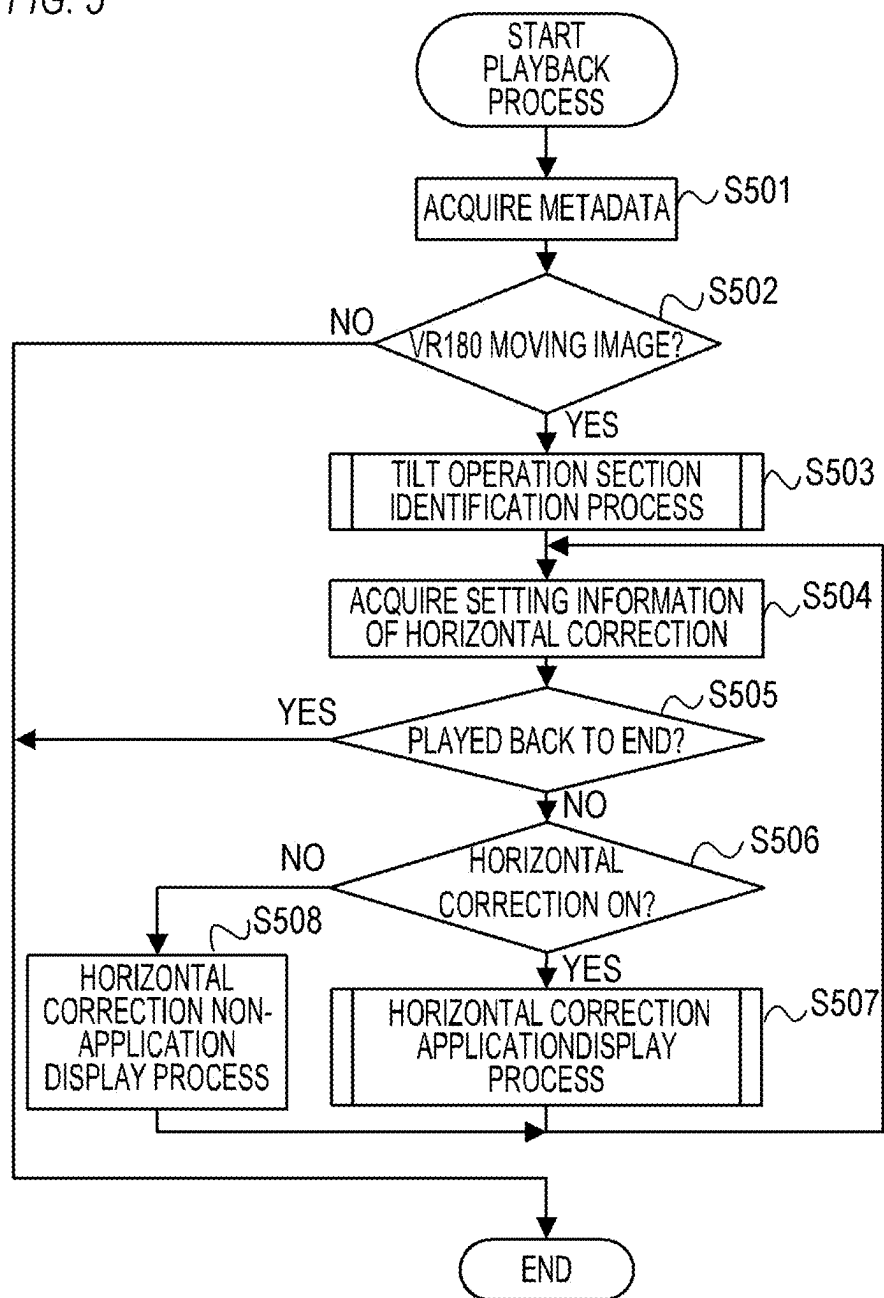


FIG. 6

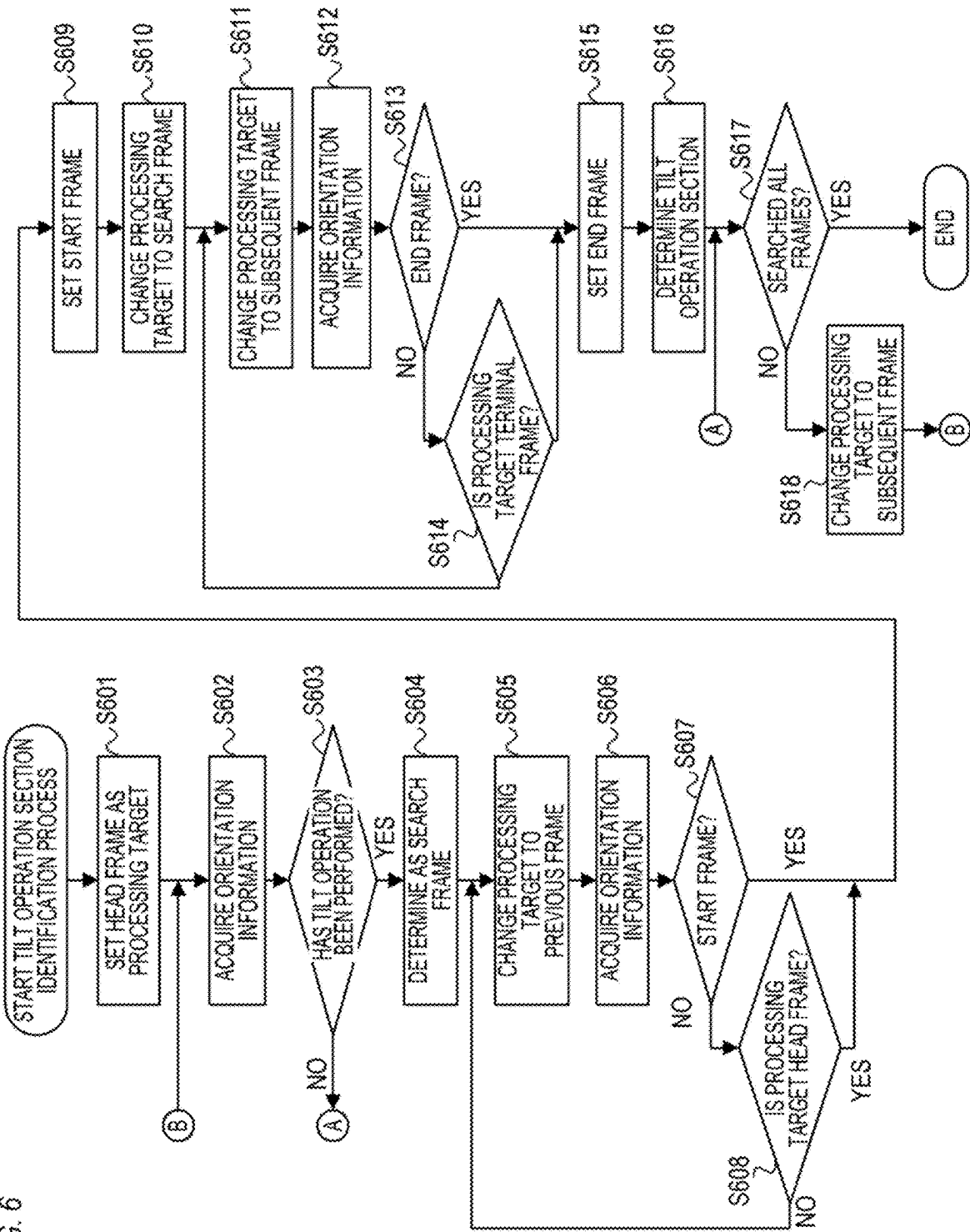


FIG. 7

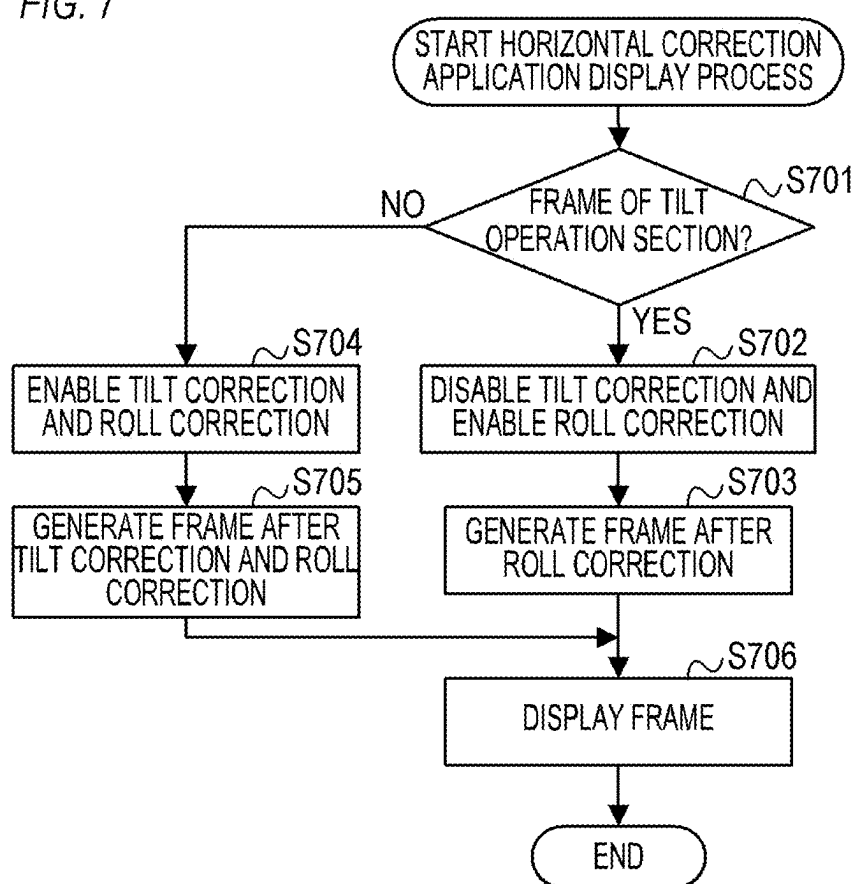


FIG. 8

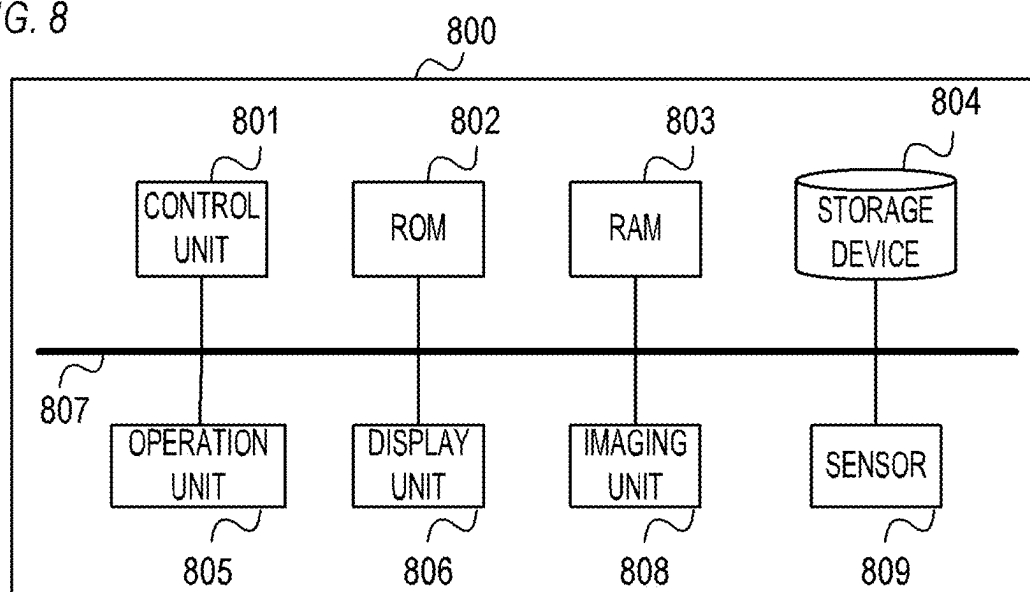


FIG. 9

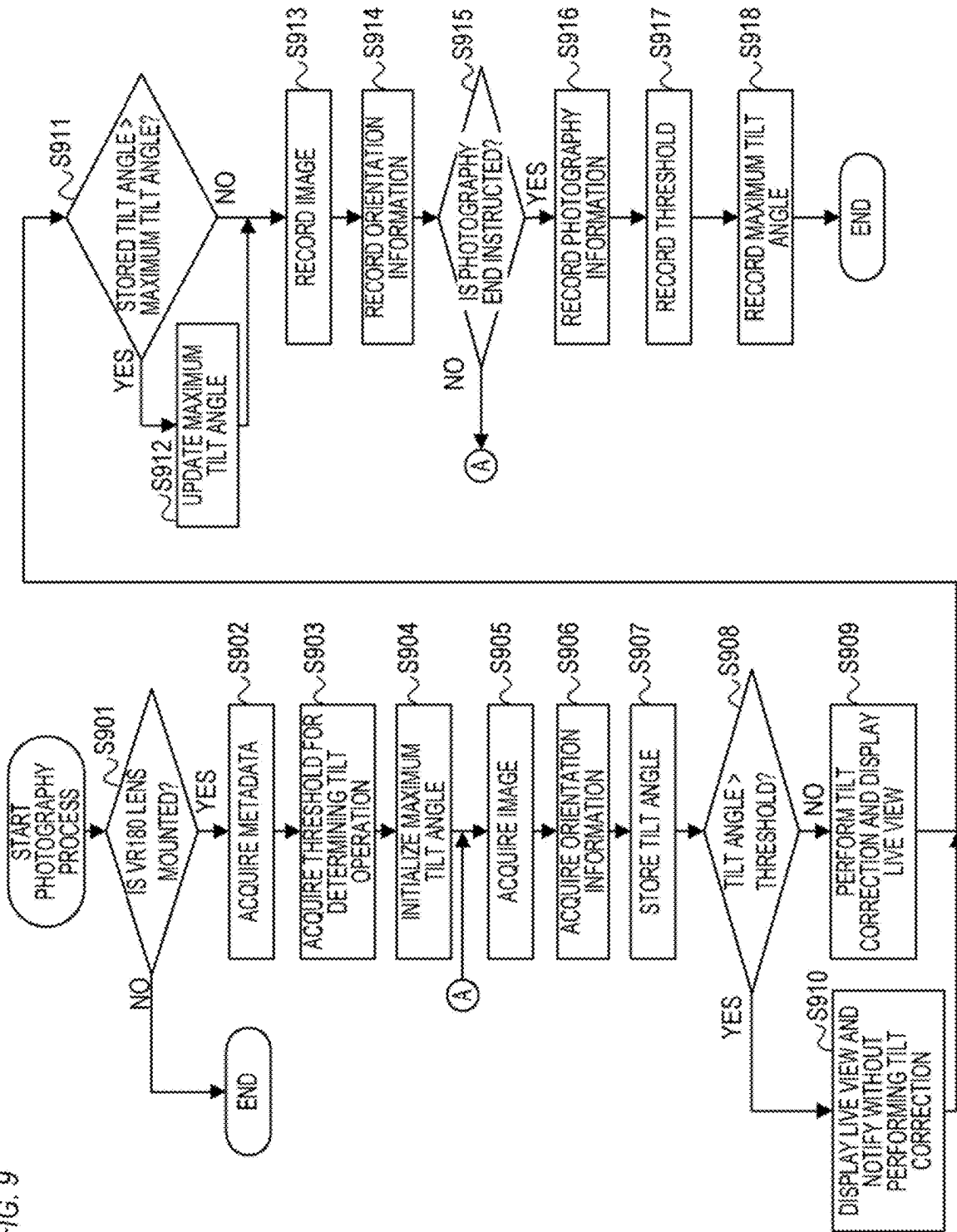


FIG. 10

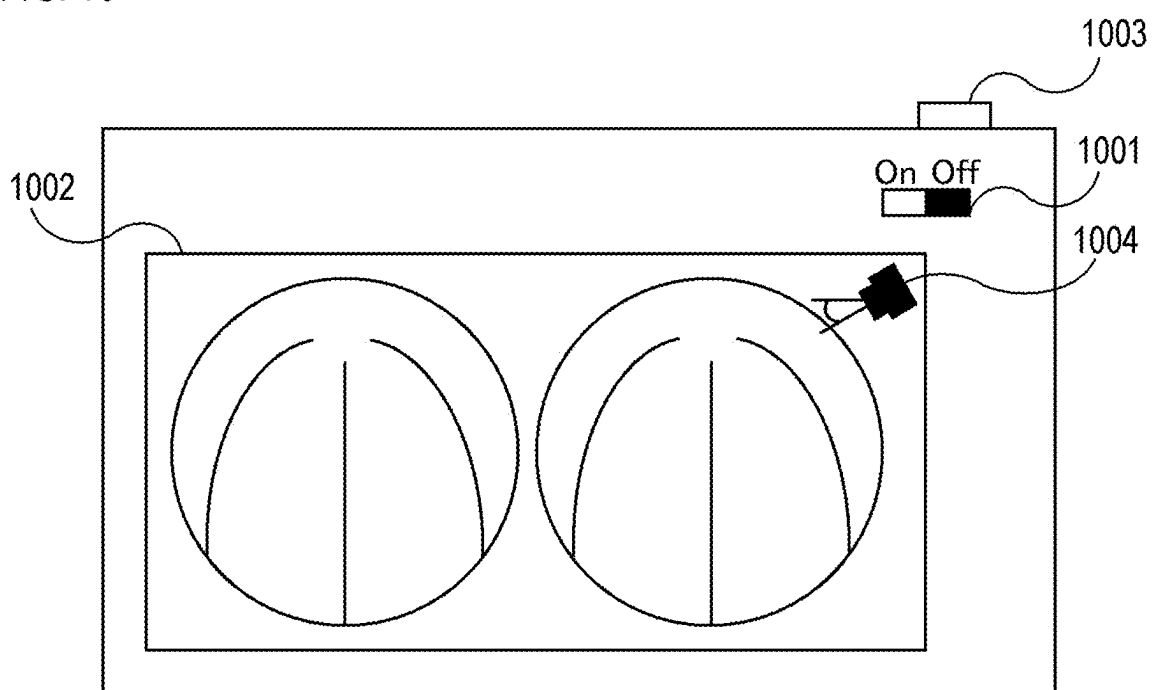


FIG. 11

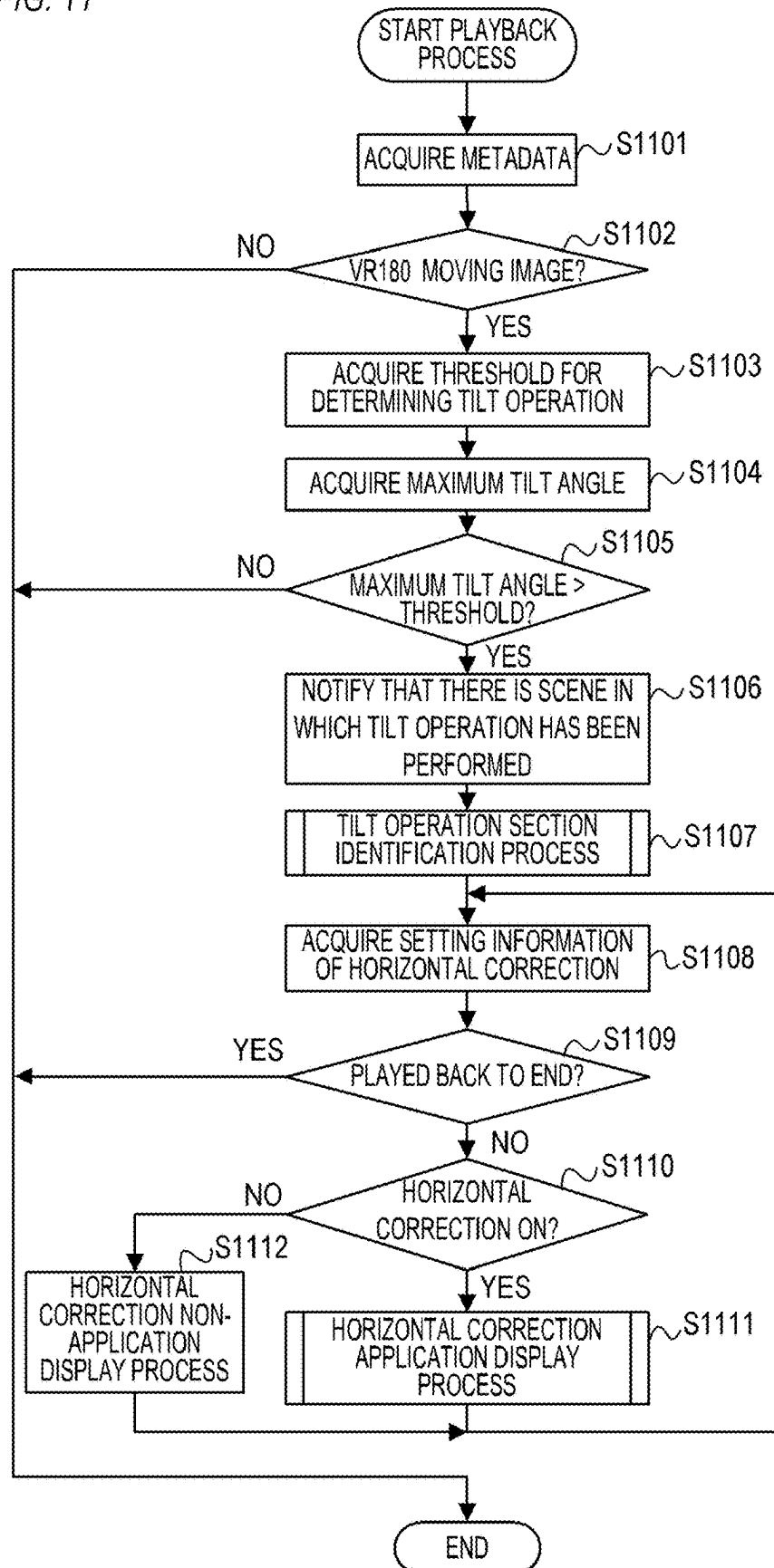


FIG. 12

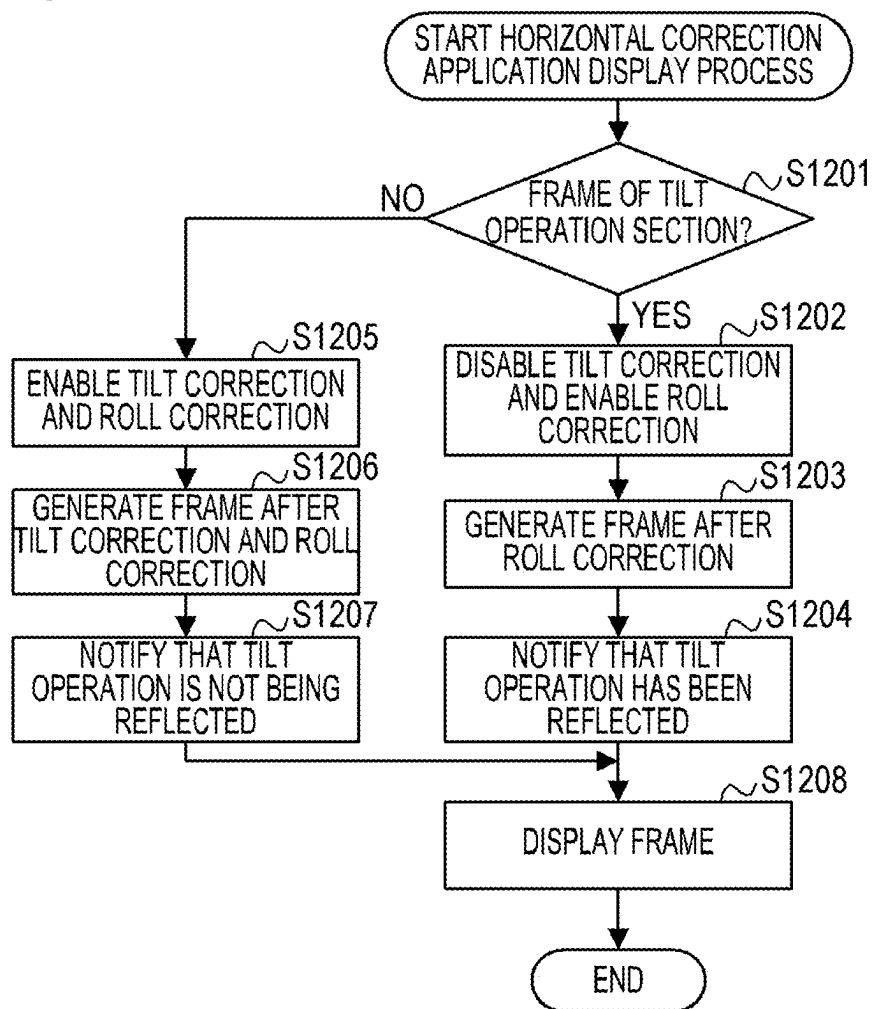
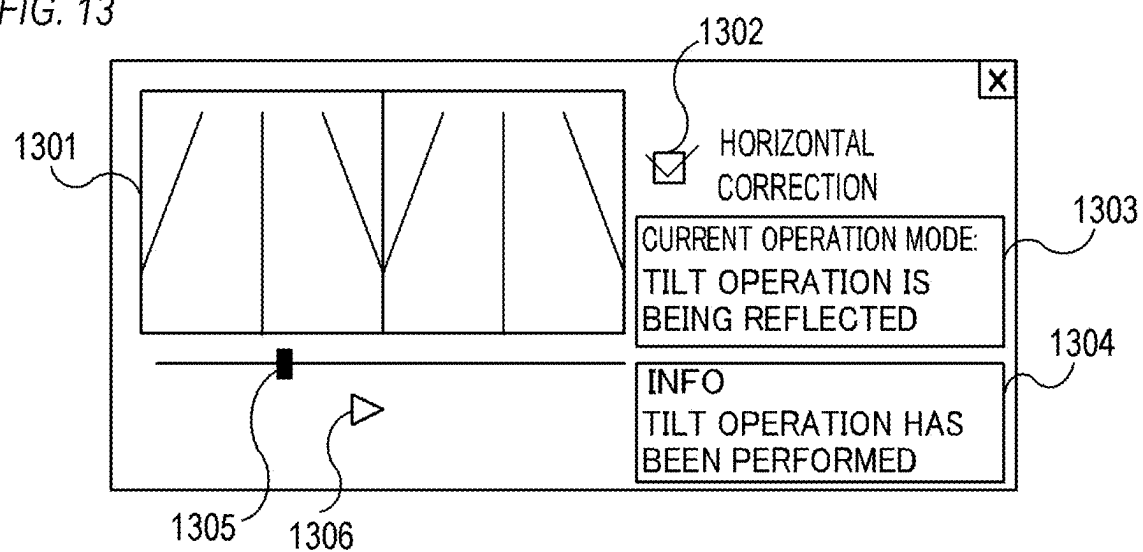


FIG. 13



ELECTRONIC DEVICE, CONTROL METHOD OF ELECTRONIC DEVICE, AND NON-TRANSITORY COMPUTER READABLE MEDIUM

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an electronic device, a control method of an electronic device, and a non-transitory computer readable medium.

Description of the Related Art

[0002] There is a camera capable of photographing one image including two parallax image areas using two optical systems facing the same direction (VR180 lens, dual lens). When the photographed image is divided into an image for right eye and an image for left eye and displayed on a head-mounted display (HMD), a user can stereoscopically view a VR image (for example, a VR180 image).

[0003] A front direction of the user when viewing the VR image is usually a horizontal direction. Therefore, based on orientation information of the camera during photography, horizontal correction is performed to reduce deviation in the horizontal direction (a tilt direction and a roll direction) regarding the front direction in the VR image. As there is a possibility that camera shake (blurring in a photographing direction) occurs when a moving image is photographed by a handheld device, horizontal correction is always performed.

[0004] JP 2012-90259 A discloses a technique of performing control according to a change in a photographing direction. In the technique disclosed in JP 2012-90259 A, moving image photography can be switched between 3D video photography and 2D video photography according to whether a tilt operation of inclining the camera in a tilt direction is performed.

[0005] However, when horizontal correction is always performed, a video on which an intentional tilt operation of a photographer is reflected (a video in which the horizontal direction is intentionally shifted from the front direction) cannot be photographed. The problem cannot be solved even when the technique disclosed in JP 2012-90259 A is used.

SUMMARY OF THE INVENTION

[0006] The present invention provides a technique capable of achieving both horizontal correction and reflection of intentional tilt operation.

[0007] The present invention in its first aspect provides an electronic device including: a processor; and a memory storing a program which, when executed by the processor, causes the electronic device to execute acquisition processing of acquiring information on orientation of an imaging apparatus during photography of an image; and execute control processing of controlling, based on the information, whether to perform tilt correction on the image to bring a position corresponding to a horizontal direction during the photography closer to a position corresponding to a front direction.

[0008] The present invention in its second aspect provides an electronic device including: a processor; and a memory storing a program which, when executed by the processor, causes the electronic device to execute acquisition process-

ing of acquiring information on orientation of an imaging apparatus; and execute control processing of performing control to provide, based on the information, a second notification regarding whether to perform tilt correction on an image photographed by the imaging apparatus to bring a position corresponding to a horizontal direction close to a position corresponding to a front direction.

[0009] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIGS. 1A and 1B are schematic diagrams describing horizontal correction in the related art;

[0011] FIG. 2 is a block diagram illustrating a configuration of a playback device according to a first embodiment;

[0012] FIG. 3 is a schematic diagram of a structure of a moving image file;

[0013] FIGS. 4A, 4B, and 4C are schematic diagrams describing horizontal correction according to the first embodiment;

[0014] FIG. 5 is a flowchart illustrating a playback process according to the first embodiment;

[0015] FIG. 6 is a flowchart illustrating a tilt operation section identification process;

[0016] FIG. 7 is a flowchart illustrating a horizontal correction application display process according to the first embodiment;

[0017] FIG. 8 is a block diagram illustrating a configuration of a camera according to a second embodiment;

[0018] FIG. 9 is a flowchart illustrating a photography process according to the second embodiment;

[0019] FIG. 10 is a schematic diagram of the camera according to the second embodiment as viewed from a back surface;

[0020] FIG. 11 is a flowchart illustrating a playback process according to the second embodiment;

[0021] FIG. 12 is a flowchart illustrating a horizontal correction application display process according to the second embodiment; and

[0022] FIG. 13 is a schematic diagram illustrating a UI screen according to the second embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0023] Hereinafter, embodiments of the present invention will be described with reference to the accompanying drawings.

First Embodiment

[0024] When photographing in VR180, a lens unit having two optical systems facing the same direction (dual-lens unit, VR180 lens unit) is used, VR180 being one of formats of VR images capable of dual stereoscopic vision. A circumferential fisheye lens capable of capturing a range of approximately 180 degrees is used for each optical system of the dual-lens unit. A front direction of the user when viewing the VR image is usually a horizontal direction. Therefore, an image photographed by the camera (imaging apparatus) equipped with the dual-lens unit is subjected to horizontal correction of bringing a position corresponding to the horizontal direction during photography of an image close to a position corresponding to the front direction. In horizontal correction, correction in a tilt direction is referred to as tilt

correction, and correction in a roll direction is referred to as roll correction. The front direction of the user is, for example, a direction of the user wearing the HMD when the user faces the front. A front direction of the image is, for example, a direction of a display range displayed when the user wearing the HMD faces the front, or a direction of the display range displayed on the HMD at first.

[0025] Since the front direction of the user when viewing the VR image is usually the horizontal direction, a photographer holds the camera so that two optical systems of the dual-lens unit are arranged horizontally and photographs the VR image. Therefore, it is less likely that the photographer performs photography by largely inclining the camera in the roll direction. However, the photographer may perform photography by inclining the camera in the tilt direction. For example, when the photographer desires to photograph an image of a road (ground), the photographer performs photography by largely inclining the camera in the tilt direction.

[0026] FIGS. 1A and 1B are schematic diagrams of a UI screen describing horizontal correction in the related art. Images 102 and 112 are images of a road photographed by performing tilt operation on a camera equipped with a dual-lens unit downward (tilt photographed). The images 102 and 112 are images after equirectangular transformation is performed on circumferential fisheye images. Check boxes 101 and 111 are check boxes for instructing whether to perform horizontal correction. Since the check box 101 is not checked, horizontal correction is not performed on the image 102. Since the check box 111 is checked, horizontal correction is performed on the image 112 compared to the image 102. The image 112 is an image after horizontal correction.

[0027] In the image 102, a horizon 103 is displayed at the upper side, and in the image 112, a horizon 113 is brought close to a center by horizontal correction. Since an angle of view of the dual-lens unit is not 360 degrees, depending on a tilt angle during photography, an invalid area 114 without video is included in the image 112 after horizontal correction is performed. Further, due to horizontal correction, the image 112 in which tilt operation is canceled is displayed instead of the image 102 reflecting tilt operation (operation of intentionally shifting the horizontal direction from the front direction, intentional tilt operation) of the photographer. Therefore, in a first embodiment, by determining whether to perform tilt correction based on orientation information of the camera during photography, both horizontal correction and reflection of intentional tilt operation are achieved.

[0028] FIG. 2 is a block diagram illustrating an example of a configuration of a playback device (electronic device) 200 according to the first embodiment. The playback device 200 is an example of an image processing apparatus, and is, for example, a PC. The playback device 200 includes a control unit 201, a ROM 202, a RAM 203, a storage device 204, an operation unit 205, a display unit 206, and a system bus 207.

[0029] The control unit 201 controls the entire playback device 200. A control unit 201 is, for example, a central processing unit (CPU). The control unit 201 is made of at least one processor and/or at least one circuit. That is, the control unit 201 may be a processor, may be a circuit, or may be a combination of processor and circuit. The control unit 201 executes a program stored in the ROM 202 to implement each process of a flowchart to be described later.

[0030] The read only memory (ROM) 202 stores programs and parameters that do not need to be changed. Here, the program includes programs for performing flowcharts described below.

[0031] The random access memory (RAM) 203 temporarily stores constants and variables for operation of the control unit 201, programs and data supplied from the ROM 202 and external devices, and the like.

[0032] The storage device 204 stores moving image files and the like. For example, the storage device 204 may be a hard disk or a flash memory built in the playback device 200. The storage device 204 may be an optical disk detachable from the playback device 200 such as a floppy disk (FD) or a compact disk (CD), a magnetic card, an optical card, an IC card, or a memory card.

[0033] The operation unit 205 receives an operation from the user (user operation). For example, the operation unit 205 is a button or a touch panel. The operation unit 205 is used to input various operation instructions to the control unit 201.

[0034] The display unit 206 displays, for example, data stored in the playback device 200 or data supplied from external devices, and the like.

[0035] The system bus 207 communicably connects a plurality of functional units of the playback device 200 to each other.

[0036] FIG. 3 is a schematic diagram illustrating an example of a structure of a moving image file 300 handled by the playback device 200. The moving image file 300 illustrated in FIG. 3 includes a header 301, photography information 302, video data 303, audio data 304, and timed meta (timed metadata) 305.

[0037] In the header 301, information indicating a type or a structure of the moving image, such as whether the moving image is a VR180 moving image, is recorded. In the photography information 302, information such as a date and time of photography, a length (time) of a photographed moving image, a camera name and a lens name used for photography, and settings during photography is recorded. Photographed moving image data is recorded in the video data 303. Audio data of the photographed moving image is recorded in the audio data 304. Metadata such as orientation information of the camera for each frame is recorded in the timed meta 305. A tilt angle and a roll angle of the camera during photography are determined based on the orientation information.

[0038] FIGS. 4A to 4C are schematic diagrams describing horizontal correction according to the first embodiment. FIGS. 4A and 4B illustrate an example in which an image of a road photographed by a camera equipped with a dual-lens unit is illustrated by equirectangular projection. It is assumed that a left-right difference between a right image area and a left image area photographed by each optical system of the dual-lens unit is corrected, and only one of the right image area and the left image area is illustrated. FIG. 4C illustrates the tilt angle of the camera.

[0039] Frames 401 to 405 in FIG. 4A are frames after horizontal correction in the related art is performed, and tilt correction and roll correction are uniformly performed on all frames.

[0040] The frame 401 is a frame photographed by holding the camera horizontally. The frame 402 is a frame photographed by tilt-operating the camera downward, and a horizon in the frame 402 is shown at the same position as a

horizon in the frame **401** by horizontal correction. The frame **403** is a frame photographed by further tilt-operating the camera downward than the frame **402**. Since the tilt angle is large, an invalid area **407** without video exists in the frame **403**. The frame **404** is a frame photographed by tilt-operating the camera downward by a similar degree as the frame **402**. The frame **405** is a frame photographed by holding the camera horizontally. A line **406** indicates positions of horizons in the frames **401** to **405**.

[0041] Frames **411** to **415** in FIG. 4B are frames after horizontal correction according to the first embodiment is performed. The frames **411** to **415** correspond to the frames **401** to **405**, respectively. In FIG. 4A, tilt correction is performed for all frames, but in FIG. 4B, it is determined whether to perform tilt correction for each frame, and tilt correction is not performed for the frames **412** to **414**. Note that, also in FIG. 4B, roll correction is performed for all frames.

[0042] A line **416** indicates positions of horizons in the frame **411** and the frame **415**. A line **417** indicates positions of horizons in the frame **412** and the frame **414**. A line **418** indicates a position of a horizon in the frame **413**.

[0043] FIG. 4C is a schematic diagram illustrating a tilt angle of the camera during photography of the frames **401** to **405** and **411** to **415**.

[0044] A line **421** indicates a direction of the camera during photography of the frames **401** and **411**. Since the frames **401** and **411** are photographed while the camera is held horizontally, the tilt angle here is 0 degrees.

[0045] An angle **422** indicates a tilt angle of the camera during photography of the frames **402** and **412**. Since the frames **402** and **412** are frames immediately after starting the tilt operation, the angle **422** is smaller than an angle **423**.

[0046] The angle **423** indicates a tilt angle of the camera during photography of the frames **403** and **413**. The angle **423** is the largest of angles **422** to **424**.

[0047] The angle **424** indicates a tilt angle of the camera during photography of the frames **404** and **414**. Since the frame **404** and **414** are frames immediately before ending the tilt operation, the angle **424** is smaller than the angle **423**.

[0048] A line **425** indicates a direction of the camera during photography of the frames **405** and **415**. Since the frames **405** and **415** are photographed while the camera is held horizontally, the tilt angle here is 0 degrees.

[0049] As illustrated in the frames **402** to **404**, the tilt operation is not reflected in the horizontal correction in the related art, but as illustrated in the frames **412** to **414**, the tilt operation is reflected in the horizontal correction according to the first embodiment. Next, details of the horizontal correction according to the first embodiment will be described with reference to FIGS. 5 to 7.

[0050] FIG. 5 is a flowchart illustrating an example of a playback process of the playback device **200**. The playback process is realized by the control unit **201** loading a program recorded in the ROM **202** to the RAM **203** and executing the program. For example, when the user operates the operation unit **205** to select the moving image file **300** and give a playback instruction, the playback process of FIG. 5 starts. Note that the playback instruction is an instruction performed on a moving image playback software, and includes, for example, a playback instruction to start playback from beginning of the moving image, a seek instruction to start playback from midway of the moving image, and the like.

[0051] In step **S501**, the control unit **201** acquires meta-data of the moving image file **300**. For example, the control unit **201** acquires data recorded in the header **301**, the photography information **302**, and the timed meta **305** illustrated in FIG. 3.

[0052] In step **S502**, the control unit **201** determines whether the moving image selected by the user is a VR180 moving image from the metadata acquired in step **S501**. When the moving image selected by the user is a VR180 moving image, the process proceeds to step **S503**, and otherwise, the processing flow ends.

[0053] In step **S503**, the control unit **201** performs processing of specifying a section in which the tilt operation has been performed (tilt operation section identification process) based on the orientation information recorded in the timed meta **305**. The tilt operation section identification process will be described later with reference to FIG. 6.

[0054] In step **S504**, the control unit **201** acquires setting information of horizontal correction. For example, the user can set whether to enable or disable the horizontal correction by checking or unchecking the check box **101** on the moving image playback software in which the UI screen as illustrated in FIGS. 1A and 1B is displayed. When the check box **101** is checked, the control unit **201** acquires information indicating that setting of the horizontal correction is enabled (horizontal correction On). When the check box **101** is unchecked, the control unit **201** acquires information indicating that the horizontal correction is disabled (horizontal correction Off).

[0055] In step **S505**, the control unit **201** determines whether the moving image has been played back to the end. When the moving image has been played back to the end, the processing flow ends, and otherwise, the process proceeds to step **S506**. Note that the determination in step **S505** may be performed before step **S504**.

[0056] In step **S506**, the control unit **201** determines whether the horizontal correction On is acquired in step **S504**. When horizontal correction On is acquired, the process proceeds to step **S507**, and otherwise, the process proceeds to step **S508**.

[0057] In step **S507**, the control unit **201** performs a process of performing horizontal correction and displaying a frame (horizontal correction application display process). The horizontal correction application display process will be described later with reference to FIG. 7.

[0058] In step **S508**, the control unit **201** performs a process of displaying a frame without performing the horizontal correction (horizontal correction non-application display process). When there is deviation in the horizontal direction (roll direction and tilt direction) regarding the front direction in the image (frame), the control unit **201** displays the image on the display unit **206** while the deviation remains. The displayed image is an image converted by equirectangular projection.

[0059] The tilt operation section identification process in step **S503** of FIG. 5 will be described with reference to FIG. 6.

[0060] In step **S601**, the control unit **201** sets a head frame of the moving image (a first frame of the moving image) as a target frame (processing target) to perform analysis from a head of the moving image file **300**.

[0061] In step **S602**, the control unit **201** acquires orientation information of the camera during photography of the target frame from the timed meta **305**.

[0062] In step S603, the control unit 201 determines whether tilt operation has been performed during photography of the target frame. When the tilt operation has been performed, the process proceeds to step S605, and otherwise, the process proceeds to step S617.

[0063] The process of step S603 will be described in detail. The control unit 201 applies the horizontal correction (tilt correction and roll correction) to the target frame and generates an image converted by the equirectangular projection (corrected image). In the roll correction, the control unit 201 acquires a roll angle from the orientation information of the camera and rotates the target frame by the roll angle to a side opposite to a direction of rolling. In the tilt correction, the control unit 201 acquires a tilt angle from the orientation information of the camera and rotates the target frame by the tilt angle to a side opposite to a direction of tilting. As such, a position corresponding to the horizontal direction during photography of the image can be brought close to a position corresponding to the front direction. The control unit 201 acquires a size of an invalid area without video in the image after correction. The invalid area is an area such as the invalid area 407 in FIG. 4A, and is an area in which the pixel values of RGB are all 0. Then, the control unit 201 acquires a ratio of the invalid area to the entire image after correction. The control unit 201 determines whether a ratio of the invalid area to the entire image after correction is larger than a threshold Th1 (for example, 10%). When the ratio is larger than the threshold Th1, the control unit 201 determines that the tilt operation has been performed during photography of the target frame, and otherwise, the control unit determines that the tilt operation has been not performed. In the example of FIG. 4A, the control unit 201 determines that the tilt operation is performed on the frame 403. The reason is that the ratio of the invalid area 407 to the entire frame 403 is larger than the threshold Th1.

[0064] Here, it is determined whether the tilt operation has been performed by determining whether the ratio of the invalid area is larger than the threshold Th1, but a method of determining whether the tilt operation has been performed is not limited thereto. For example, it may be determined whether the tilt operation has been performed by determining whether a tilt amount (tilt angle) is larger than a threshold or whether a size (for example, the number of pixels) of the invalid area is larger than a threshold. In addition, it may be determined whether the tilt operation has been performed by determining whether a size of an area (a ratio of the area to the entire image) not mapped to an output destination during calculation processing procedure of the equirectangular projection is larger than a threshold.

[0065] In step S604, the control unit 201 determines the current target frame as a search frame. The search frame is a frame that serves as a starting point for checking a frame from which the tilt operation started (start frame of the tilt operation) before the current target frame (search frame).

[0066] In step S605, the control unit 201 changes the target frame to a previous frame.

[0067] In step S606, the control unit 201 acquires orientation information of the camera during photography of the target frame from the timed meta 305.

[0068] In step S607, the control unit 201 determines whether the target frame is the start frame of the tilt operation based on the orientation information of the camera acquired in step S606. For example, when a tilt angle is smaller than a threshold Th2 (for example, 1 degree), the

control unit 201 determines that the target frame is the start frame of the tilt operation. When the target frame is the start frame of the tilt operation, the process proceeds to step S609, and otherwise, the process proceeds to step S608. Note that the threshold Th2 is not limited to 1 degree, and may be, for example, 0 degrees. The control unit 201 may determine whether the target frame is the start frame of the tilt operation by determining whether a tilt amount (tilt angle) is larger than a threshold or whether a size of the invalid area is larger than a threshold. The threshold used for the determination in step S607 corresponds to a tilt amount smaller than the tilt amount corresponding to the threshold Th1 used for the determination in step S603.

[0069] In step S608, the control unit 201 determines whether the target frame is the head frame. When the target frame is the head frame, the process proceeds to step S609, and otherwise, the process proceeds to step S605.

[0070] In step S609, the control unit 201 sets the current target frame as the start frame of the tilt operation.

[0071] By repeating the process from step S605 to step S608, the control unit 201 searches for a frame that is the start frame of the tilt operation. In the example of FIG. 4A, the control unit 201 changes the target frame from the frame 403 to the frame 402, and searches for a frame whose tilt angle is smaller than the threshold Th2 toward the head of the moving image file 300. Since the angle 422 is smaller than the threshold Th2, in step S609, the control unit 201 sets the frame 402 as the start frame of the tilt operation. The frame 402 is a frame closest to the frame 403 among frames photographed before the frame 403 while the tilt angle is smaller than the threshold Th2. When there is no frame photographed before the frame 403 while the tilt angle is smaller than the threshold Th2, in step S609, the control unit 201 sets the head frame as the start frame of the tilt operation.

[0072] In step S610, the control unit 201 reads the search frame determined in step S604 and sets the search frame as the target frame.

[0073] In step S611, the control unit 201 changes the target frame to a subsequent (next) frame.

[0074] In step S612, the control unit 201 acquires orientation information of the camera during photography of the target frame from the timed meta 305.

[0075] In step S613, the control unit 201 determines whether the target frame is an end frame of the tilt operation based on the orientation information of the camera acquired in step S612. For example, when a tilt angle is smaller than a threshold Th3 (for example, 1 degree), the control unit 201 determines that the target frame is the end frame of the tilt operation. When the target frame is the end frame of the tilt operation, the process proceeds to step S615, and otherwise, the process proceeds to step S614. The threshold Th3 and the threshold Th2 used for the determination in step S607 may be the same value or different values.

[0076] In step S614, the control unit 201 determines whether the target frame is a terminal frame (a last frame of the moving image). When the target frame is the terminal frame, the process proceeds to step S615, and otherwise, the process proceeds to step S611.

[0077] In step S615, the control unit 201 sets the current target frame as the end frame of the tilt operation.

[0078] By repeating the process from step S611 to step S614, the control unit 201 searches for a frame that is the end frame of the tilt operation. In the example of FIG. 4A, the

control unit **201** changes the target frame from the frame **403** to the frame **404**, and searches for a frame whose tilt angle is smaller than the threshold Th3 toward a terminal of the moving image file **300**. Since the angle **424** is smaller than the threshold Th3, in step S615, the control unit **201** sets the frame **404** as the end frame of the tilt operation. The frame **404** is the frame closest to the frame **403** among frames photographed after the frame **403** while the tilt angle is smaller than the threshold Th3. When there is no frame photographed after the frame **403** while the tilt angle is smaller than the threshold Th3, in step S615, the control unit **201** sets the terminal frame as the end frame of the tilt operation.

[0079] In step S616, the control unit **201** determines a period from the start frame set in step S609 to the end frame set in step S615 as a tilt operation section (identified period).

[0080] In step S617, the control unit **201** determines whether all frames of the moving image file **300** have been searched. When all frames of the moving image file **300** have been searched, the processing flow ends, and otherwise, the process proceeds to step S618.

[0081] In step S618, the control unit **201** changes the target frame to a subsequent frame.

[0082] The tilt operation section determined by the processing flow is used to determine whether the tilt operation section is a section in which an operation of the horizontal correction is to be changed in the processing flow of FIG. 7. When there are a plurality of tilt operation sections, the control unit **201** stores the plurality of sections.

[0083] The horizontal correction application display process in step S507 of FIG. 5 will be described with reference to FIG. 7.

[0084] In step S701, the control unit **201** determines whether a frame to be displayed is a frame corresponding to the tilt operation section. When the frame to be displayed is a frame corresponding to the tilt operation section, the process proceeds to step S702, and otherwise, the process proceeds to step S704.

[0085] In step S702, the control unit **201** disables the tilt correction. In a normal horizontal correction, the control unit **201** enables both the tilt correction and the roll correction, but in step S702, the operation of the horizontal correction is changed so that the tilt correction is disabled and only the roll correction is enabled.

[0086] In step S703, the control unit **201** generates a frame subjected to the roll correction (frame after roll correction).

[0087] In step S704, the control unit **201** enables the tilt correction and the roll correction.

[0088] In step S705, the control unit **201** generates a frame on which the tilt correction and the roll correction have been performed (frame after tilt correction and roll correction).

[0089] In step S706, the control unit **201** displays the frame generated in step S703 or step S705 and ends the processing flow.

[0090] As such, the control unit **201** determines the ratio of the invalid area in the image after the horizontal correction (roll correction and tilt correction) is performed to the entire image and controls not to perform the tilt correction when the ratio is larger than the threshold Th1. When the moving image includes a period to which it is determined that the tilt correction is not to be performed (a period in which the above ratio is larger than the threshold Th1), the control unit **201** performs control not to perform the tilt correction in an identified period including the period. In the

example of FIG. 4B, the control unit **201** performs control not to perform the tilt correction in the period of the frames **412** to **414** that includes the frame **413** to which it is determined that the tilt correction is not to be performed. As a result, an image reflecting the tilt operation can be displayed. Further, it is possible to reduce display of an image including an invalid area. In the frames **411** and **415** that do not correspond to the tilt operation section, the tilt correction is performed, and the image in the horizontal direction is displayed when the user faces the front.

[0091] The control unit **201** may acquire information on orientation of the camera during photography and information on at least one of a type and an angle of view of a lens unit mounted on the camera during photography, and may determine whether to perform the tilt correction. For example, the threshold Th1 for determining whether to perform tilt correction may be determined based on the angle of view of the lens unit. The threshold Th1 may be set to 15% when the angle of view of the lens unit is 190 degrees and may be set to 5% when the angle of view of the lens unit is 170 degrees. Further, the thresholds Th1 to Th3 may be set by the user.

[0092] Although a case in which the processing target is the moving image file **300** has been described, the processing target may be a still image file. When metadata similar to that of the moving image file **300** is attached to the still image file, a process similar to the process performed in the moving image file **300** can be executed by continuously processing a plurality of still image files.

[0093] As described above, in the first embodiment, when intentional tilt operation is performed, the control unit **201** controls not to perform the tilt correction, and otherwise, the control unit **201** controls to perform the tilt correction. Accordingly, it is possible to achieve both horizontal correction and reflection of intentional tilt operation.

Second Embodiment

[0094] In the first embodiment, a threshold for determining whether to perform tilt correction is a fixed value that does not depend on a lens used for photography. In the second embodiment, a threshold for determining whether to perform tilt correction is changed depending on a lens to be used for photography. Further, notification regarding whether to perform tilt correction is performed. Hereinafter, differences from the first embodiment will be mainly described.

[0095] FIG. 8 is a block diagram illustrating an example of a configuration of a camera **800** according to the second embodiment. The camera **800** is an example of an image processing apparatus, and includes a control unit **801**, a ROM **802**, a RAM **803**, a storage device **804**, an operation unit **805**, a display unit **806**, a system bus **807**, an imaging unit **808**, and a sensor **809**. Functions of the control unit **801**, the ROM **802**, the RAM **803**, and the storage device **804** are similar to the functions of the control unit **201**, the ROM **202**, the RAM **203**, and the storage device **204** in the playback device **200** of FIG. 2. Further, functions of the operation unit **805**, the display unit **806**, and the system bus **807** are similar to the functions of the operation unit **205**, the display unit **206**, and the system bus **207** in the playback device **200**.

[0096] The imaging unit **808** is, for example, a charge coupled device (CCD) and inputs a video signal.

[0097] The sensor **809** is a sensor for acquiring orientation information of the camera **800**. For example, an acceleration sensor, a gyro sensor, or the like can be used for the sensor **809**.

[0098] It is assumed that a structure of a moving image file recorded by the camera **800** is the same as that in the example of FIG. 3, and that the playback device **200** of FIG. 2 plays back the moving image file recorded by the camera **800**.

[0099] FIG. 9 is a flowchart illustrating an example of a photography process of the camera **800**. The photography process is realized by the control unit **801** loading a program recorded in the ROM **802** to the RAM **803** and executing the program. For example, when the user operates the operation unit **805** to give an instruction to start photography, the photography process in FIG. 9 starts.

[0100] In step S901, the control unit **801** determines whether the mounted lens is a VR180 lens. When the mounted lens is a VR180 lens, the process proceeds to step S902, and otherwise, the processing flow ends.

[0101] In step S902, the control unit **801** acquires meta-data indicating a lens name, an angle of view of the lens, an adjustment value of optical correction, and the like from the VR180 lens.

[0102] In step S903, the control unit **801** acquires a threshold Th4 for determining the tilt operation. The threshold Th4 is a threshold for determining whether a tilt operation has been performed (tilt correction is not to be performed) when a tilt amount (for example, tilt angle) is larger than a value of the threshold Th4. When the threshold Th4 is recorded in the lens, the control unit **801** acquires the threshold Th4 from the lens. When the threshold Th4 determined for each lens is recorded in the ROM **802**, the control unit **801** acquires the threshold Th4 corresponding to the mounted lens from the ROM **802**. The threshold Th4 may be a value that can be set by the user. The threshold Th4 is determined based on, for example, the angle of view of the lens.

[0103] In step S904, the control unit **801** initializes a stored maximum tilt angle to 0 degrees.

[0104] In step S905, the control unit **801** acquires an image (live image) captured by the imaging unit **808**.

[0105] In step S906, the control unit **801** acquires current orientation information of the camera **800** output from the sensor **809**.

[0106] In step S907, the control unit **801** stores a tilt angle indicated by the orientation information acquired in step S906 as an absolute value. The reason why the tilt angle is stored as an absolute value is that up-and-down directions are not differentiated when determining whether the tilt operation has been performed.

[0107] In step S908, the control unit **801** determines whether the current tilt angle stored in step S907 is larger than the threshold Th4 acquired in step S903. When the current tilt angle is larger than the threshold Th4, the process proceeds to step S910, and otherwise, the process proceeds to step S909.

[0108] In step S909, the control unit **801** displays (in live view) the image captured by the imaging unit **808** and subjected to the tilt correction on the display unit **806**.

[0109] In step S910, the control unit **801** displays the image captured by the imaging unit **808** and not subjected to the tilt correction on the display unit **806**. Further, the control unit **801** issues a notification indicating that it is

determined that the tilt operation has been performed. An icon **1004** in FIG. 10 is an example of an icon displayed by the notification in step S910, and indicates that the tilt operation is being performed. The control unit **801** may make a notification by a method other than displaying an icon (for example, display of text, lighting of lamp, output of sound, and the like).

[0110] Note that the control unit **801** is not limited to making a notification when displaying an image without performing the tilt correction (step S910), and may make a notification when performing the tilt correction and displaying an image (step S909). For example, in step S909, the control unit **801** may make a notification indicating that it is not determined that the tilt operation has been performed or that the tilt correction has been performed. The roll correction is performed in both step S909 and step S910.

[0111] In step S911, the control unit **801** determines whether the tilt angle stored in step S907 is larger than the maximum tilt angle. The maximum tilt angle is the value initialized in step S904 (0 degrees) when the maximum value of the tilt angle is not updated in step S912, and is the updated value when the maximum value of the tilt angle is updated in step S912. When the tilt angle stored in step S907 is larger than the stored maximum tilt angle, the process proceeds to step S912, and otherwise, the process proceeds to step S913.

[0112] In step S912, the control unit **801** updates the maximum tilt angle to the tilt angle stored in step S907.

[0113] In step S913, the control unit **801** records the image acquired in step S905 in the video data **303**.

[0114] In step S914, the control unit **801** records the orientation information acquired in step S906 in the timed meta **305**. Note that the control unit **801** may record the orientation information in another file associated with the moving image file **300** other than the timed meta **305**.

[0115] In step S915, the control unit **801** determines whether the user operated the operation unit **805** (for example, presses a button **1003** in FIG. 10) to give an instruction to end photography. When the instruction to end photography is given, the process proceeds to step S916, and otherwise, the process proceeds to step S905. The control unit **801** repeats the process from step S905 to step S915 until the instruction to end photography is given in step S915.

[0116] In step S916, the control unit **801** records information such as a date and time of photography and a camera name and a lens name used in the photography in the photography information **302**.

[0117] In step S917, the control unit **801** records the threshold used in the photography (the threshold Th4 acquired in step S903) as a part of the metadata of the moving image file **300**.

[0118] In step S918, the control unit **801** records the stored maximum tilt angle as a part of the metadata of the moving image file **300**.

[0119] By reading the values recorded in steps S916 to S918 during playback of the moving image, it is possible to easily determine whether the tilt operation has been performed during photography of the moving image by using the threshold Th4 corresponding to the lens used for the photography of the moving image. Note that the maximum tilt angle may be calculated by searching all frames during playback of the moving image.

[0120] Although an example in which the image before horizontal correction is always recorded has been described, there may be other examples. For example, the image displayed on the display unit **806** (the image displayed in step **S909** or **S910**) may be recorded. Then, the horizontal correction (tilt correction or roll correction) during playback becomes unnecessary.

[0121] FIG. **10** is a schematic diagram of the camera **800** as viewed from the back surface (a side opposite to an object side). A switch **1001** is a power switch that switches between On and Off of a power of the camera **800**, and is an example of the operation unit **805**. A display unit **1002** is an example of the display unit **806** that displays a live view image of the camera **800**. A button **1003** is a shutter button for giving a photography instruction of a still image or a moving image, and is an example of the operation unit **805**. The icon **1004** is an example of the icon notified in step **S910** indicating that the tilt operation is being performed. The user can recognize that the tilt operation is being performed by looking at the icon **1004**.

[0122] FIG. **11** is a flowchart illustrating an example of a playback process of the playback device **200** according to the second embodiment. The playback process is realized by the control unit **201** loading a program recorded in the ROM **202** to the RAM **203** and executing the program. For example, when the user operates the operation unit **205** to select the moving image file and give a playback instruction, the playback process of FIG. **11** starts. Steps **S1101**, **S1102**, and **S1107** to **S1112** are the same as steps **S501**, **S502**, and **S503** to **S508** of FIG. **5**. The threshold **Th1** is used in the tilt operation section identification process of the first embodiment, but the threshold **Th4** is used in the tilt operation section identification process of the second embodiment. Note that the control unit **201** may perform the tilt operation section identification process during playback of the moving image by using a threshold different from the threshold **Th4** used during photography of the moving image. Further, information indicating the tilt operation section may be recorded during photography, and the control unit **201** may perform control not to perform the tilt correction in a period indicated by the information.

[0123] In step **S1103**, the control unit **201** acquires the threshold **Th4** recorded in step **S917** of FIG. **9** from the moving image file **300**.

[0124] In step **S1104**, the control unit **201** acquires the maximum tilt angle recorded in step **S918** from the moving image file **300**.

[0125] In step **S1105**, the control unit **201** determines whether the maximum tilt angle acquired in step **S1104** is larger than the threshold **Th4** acquired in step **S1103**. When the maximum tilt angle is larger than the threshold **Th4**, the process proceeds to step **S1106**, and otherwise, the processing flow ends.

[0126] In step **S1106**, the control unit **201** makes a notification indicating that the tilt operation has been performed during photography of the playing moving image (there is a scene in which the tilt operation has been performed). For example, the control unit **201** displays a text “intentional tilt operation has been performed” in an area **1304** of FIG. **13** to indicate that the tilt operation has been performed during photography. Note that the notification is not limited to display of text, and may be performed by display of icon or output of voice.

[0127] The horizontal correction application display process in the second embodiment (step **S1111** of FIG. **11**) is different from the horizontal correction application display process in the first embodiment (step **S507** of FIG. **5**). The horizontal correction application display process in step **S1111** of FIG. **11** will be described with reference to FIG. **12**. Steps **S1201** to **S1203**, **S1205**, **S1206**, and **S1208** are the same as steps **S701** to **S706** of FIG. **7**.

[0128] In step **S1204**, the control unit **201** makes a notification indicating that the tilt operation has been reflected. For example, the control unit **201** displays text “intentional tilt operation is being reflected” in an area **1303** in FIG. **13** to indicate that the tilt operation has been reflected. As such, the control unit **201** performs control to notify whether the tilt correction is performed based on the orientation information of the camera. When a tilt amount of the camera corresponding to the orientation information is a tilt amount for which tilt correction is not performed, the control unit **201** performs control to make a notification indicating that the tilt operation has been reflected. As a result, the user can easily recognize whether the tilt correction is performed (whether the tilt operation is reflected).

[0129] In step **S1207**, the control unit **201** makes a notification indicating that the tilt operation is not being reflected (the operation of the horizontal correction is not changed). For example, the control unit **201** displays text “tilt correction and roll correction is being applied” indicating that the tilt operation is not being reflected in the area **1303**.

[0130] In steps **S1204** and **S1207**, the control unit **201** makes a notification by display of text, but may make a notification by display of icon, for example.

[0131] FIG. **13** is a schematic diagram of a UI screen displayed on the display unit **206**.

[0132] An image **1301** is an image including a plurality of image areas (a left image area and a right image area) photographed by a plurality of optical systems of a dual-lens unit. The image **1301** is an image obtained by converting circumferential fisheye images by equirectangular projection. A bar **1305** is a position bar indicating a playback position of a moving image. The user can operate the bar **1305** to display a certain frame. A button **1306** is a playback button for instructing playback of the moving image.

[0133] A check box **1302** is a check box for instructing enabling or disabling (application or non-application) of the horizontal correction.

[0134] The area **1303** is an area indicating information regarding an operation mode of the horizontal correction. When the check box **1302** is unchecked and the horizontal correction is disabled, a display indicating that the horizontal correction is disabled (display of blank, display of “horizontal correction is disabled”, and the like) is performed in the area **1303**. When the check box **1302** is checked and the horizontal correction is enabled, a display indicating an operation state of the horizontal correction (whether the tilt operation is being reflected) is performed.

[0135] The area **1304** is an area indicating whether the tilt operation has been performed during photography of the playing moving image. In the area **1304**, a display indicating whether the currently displayed frame is a frame on which the tilt operation has been performed may be performed.

[0136] The control unit **201** may give a notification indicating that the tilt operation has been performed for each moving image file when displaying a list of candidates of a

moving image file to be played back on the UI screen of the moving image playback software instead of during playback of the moving image.

[0137] In the first embodiment and the second embodiment, only a tilt component and a roll component are corrected in the horizontal correction. Even if three components of pan, tilt, and roll are correction targets of horizontal correction, when an intentional tilt operation is performed during photography of an image, the control unit **201** performs control not to perform the tilt correction on the image. Further, when an intentional pan operation is performed during photography of an image, the control unit **201** performs control not to perform pan correction (correction for reducing deviation in a panning direction regarding the front direction in the image) on the image. When an intentional operation is performed in an oblique direction during photography of an image, the control unit **201** performs control not to perform correction in the oblique direction (pan correction and tilt correction) on the image.

[0138] The control unit **201** may further perform a process of reducing blurs in a video due to camera shake during photography. For example, the control unit **201** may perform a process of reducing blurs in a video in the up-and-down directions due to camera shaking in the tilt direction during photography.

[0139] As described above, in the second embodiment, the threshold for determining whether an intentional tilt operation has been performed is changed depending on the lens used for photography. As a result, it is possible to more suitably determine (with high accuracy) whether an intentional tilt operation has been performed. Further, notification is performed when an intentional tilt operation has been performed during photography of a moving image and when a scene in which an intentional tilt operation has been performed during photography is played back. As a result, the user can easily recognize whether an intentional tilt operation has been performed.

[0140] Note that the above-described various types of control may be processing that is carried out by one piece of hardware (e.g., processor or circuit), or otherwise. Processing may be shared among a plurality of pieces of hardware (e.g., a plurality of processors, a plurality of circuits, or a combination of one or more processors and one or more circuits), thereby carrying out the control of the entire device.

[0141] Also, the above processor is a processor in the broad sense, and includes general-purpose processors and dedicated processors. Examples of general-purpose processors include a central processing unit (CPU), a micro processing unit (MPU), a digital signal processor (DSP), and so forth. Examples of dedicated processors include a graphics processing unit (GPU), an application-specific integrated circuit (ASIC), a programmable logic device (PLD), and so forth. Examples of PLDs include a field-programmable gate array (FPGA), a complex programmable logic device (CPLD), and so forth.

[0142] The embodiment described above (including variation examples) is merely an example. Any configurations obtained by suitably modifying or changing some configurations of the embodiment within the scope of the subject matter of the present invention are also included in the present invention. The present invention also includes other configurations obtained by suitably combining various features of the embodiment.

[0143] For example, it is described that one image in which two parallax image areas are arranged side by side is acquired, but the number of image areas, that is, the number of optical systems may be larger than two and an arrangement of a plurality of image areas is not particularly limited.

[0144] The present invention is not limited to a camera or a PC and is applicable to any electronic device that can handle an image including a plurality of image areas corresponding to a plurality of optical systems. For example, the present invention is applicable to a PDA, a mobile phone terminal, a portable image viewer, a printer device, a digital photo frame, a music player, a video game machine, an electronic book reader, a cloud server, and the like. The present invention is further applicable to, for example, a video player, a display device (including a projector), a tablet terminal, a smartphone, an AI speaker, a home electrical appliance device, and an on-vehicle apparatus. The invention is also applicable to a multi-view smartphone or the like with a plurality of optical systems of different types, such as a standard lens, a wide-angle lens, and a zoom lens.

Other Embodiments

[0145] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0146] According to the present invention, it is possible to achieve both horizontal correction and reflection of intentional tilt operation.

[0147] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0148] This application claims the benefit of Japanese Patent Application No. 2024-026389, filed on Feb. 26, 2024, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An electronic device comprising:

a processor; and

a memory storing a program which, when executed by the processor, causes the electronic device to

execute acquisition processing of acquiring information on orientation of an imaging apparatus during photography of an image; and

execute control processing of controlling, based on the information, whether to perform tilt correction on the image to bring a position corresponding to a horizontal direction during the photography closer to a position corresponding to a front direction.

2. The electronic device according to claim 1, wherein, in the acquisition processing, the information on the orientation of the imaging apparatus during the photography and at least one of a type and an angle of view of a lens unit mounted on the imaging apparatus during the photography are acquired.

3. The electronic device according to claim 1, wherein in the control processing, a size of an invalid area without video in an image after the tilt correction is performed is determined based on the information and control is performed not to perform the tilt correction in a case where the size is larger than a first threshold.

4. The electronic device according to claim 1, wherein in the control processing, a ratio of an invalid area without video in an image after the tilt correction is performed to entire of the image is determined based on the information and control is performed not to perform the tilt correction in a case where the ratio is larger than a second threshold.

5. The electronic device according to claim 1, wherein in the control processing, control is performed not to perform the tilt correction in a case where a tilt amount of the imaging apparatus corresponding to the information is larger than a third threshold.

6. The electronic device according to claim 1, wherein in a case where a moving image photographed by the imaging apparatus includes a period in which it is determined that the tilt correction is not to be performed, in the control processing, control is performed not to perform the tilt correction in an identified period including the period.

7. The electronic device according to claim 6, wherein a start frame of the identified period is a frame closest to the period in which it is determined that the tilt correction is not to be performed among frames photographed before the period in which it is determined that the tilt correction is not to be performed while a tilt amount of the imaging apparatus is smaller than a fourth threshold.

8. The electronic device according to claim 7, wherein the start frame of the identified period is a first frame of the moving image in a case where there is no frame photographed before the period in which it is determined that the tilt correction is not to be performed while the tilt amount of the imaging apparatus is smaller than the fourth threshold.

9. The electronic device according to claim 7, wherein an end frame of the identified period is a frame closest to the period in which it is determined that the tilt correction is not to be performed among frames photographed after the period in which it is determined that the tilt correction is not to be performed while the tilt amount of the imaging apparatus is smaller than a fifth threshold.

10. The electronic device according to claim 9, wherein the end frame of the identified period is a last frame of the moving image in a case where there is no frame photographed after the period in which it is determined that the tilt correction is not to be performed while the tilt amount of the imaging apparatus is smaller than the fifth threshold.

11. The electronic device according to claim 1, wherein in the control processing, control is performed to provide a first notification when an image on which the tilt correction is not performed is displayed.

12. The electronic device according to claim 1, wherein the image includes a plurality of image areas photographed through a plurality of optical systems, respectively.

13. The electronic device according to claim 1, wherein the program, when executed by the processor, further causes the electronic device to execute correction processing of performing the tilt correction.

14. An electronic device comprising:

a processor; and

a memory storing a program which, when executed by the processor, causes the electronic device to

execute acquisition processing of acquiring information on orientation of an imaging apparatus; and

execute control processing of performing control to provide, based on the information, a second notification regarding whether to perform tilt correction on an image photographed by the imaging apparatus to bring a position corresponding to a horizontal direction close to a position corresponding to a front direction.

15. The electronic device according to claim 14, wherein in the control processing, control is performed to provide the second notification in a case where a tilt amount of the imaging apparatus corresponding to the information is a tilt amount for which the tilt correction is not to be performed.

16. A control method of an electronic device, comprising: acquiring information on orientation of an imaging apparatus during photography of an image; and

controlling, based on the information, whether to perform tilt correction on the image to bring a position corresponding to a horizontal direction during the photography closer to a position corresponding to a front direction.

17. A control method of an electronic device, comprising: acquiring information on orientation of an imaging apparatus; and

controlling to provide, based on the information, a second notification regarding whether to perform tilt correction on an image photographed by the imaging apparatus to bring a position corresponding to a horizontal direction close to a position corresponding to a front direction.

18. A non-transitory computer readable medium that stores a program, wherein the program causes a computer to execute a control method of an electronic device, the control method comprising:

acquiring information on orientation of an imaging apparatus during photography of an image; and
controlling, based on the information, whether to perform tilt correction on the image to bring a position corresponding to a horizontal direction during the photography closer to a position corresponding to a front direction.

19. A non-transitory computer readable medium that stores a program, wherein the program causes a computer to execute a control method of an electronic device, the control method comprising:

acquiring information on orientation of an imaging apparatus; and
controlling to provide, based on the information, a second notification regarding whether to perform tilt correction on an image photographed by the imaging apparatus to bring a position corresponding to a horizontal direction close to a position corresponding to a front direction.

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